Haowen Yan

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BIOGRAPHY

I am Haowen Yan, a third-year student majoring in Computer Science and Technology at Sun Yat-Sen University. Currently, I am a research intern at Xspark AI. I am also privileged to intern at TEA Lab under the supervision of Prof. Huazhe Xu, where I worked on teleoperation. Before that, I was also fortunate to be mentored by Prof. Gang Chen, working on optimal control and UAV.

My research interests focus on **Embodied AI**, **Robotics**, **Reinforcement Learning**, **World Models**, and **Optimal Control**.

EDUCATION

Sun Yat-Sen University — Guangdong, China
Bachelor in Computer Science — Sept 2023 – Present

- **GPA:** 4.3397/5.0; **Rank:** 3/353 (top 0.85%)
- **Course Grades:**
  Control Theory (100/100), Optimization Theory (99/100), Mathematics Analysis II (98/100), Machine Learning (98/100), Data Structures (98/100), Signals and Systems (97/100), Probability and Statistics (96/100), etc.
- **Self-Study Courses based on Interest:**
  Optimal Control, Robotics, etc.

PROJECTS

Teleoperation of a robotic arm — 2026.01–2026.02
Advisors: Prof. Huazhe Xu — *Tsinghua University*

- **Introduction**: In many existing teleoperation systems, the robot motion is affected by the head-mounted display (HMD): even when the controller remains stationary, movement of the HMD alone can cause unintended robot motion.
- **Involvement**:
  - Implemented teleoperation of an existing robotic arm using Meta Quest 3S based on an incremental control scheme.
  - By introducing the initial pose of the HMD as a reference frame, this work eliminates the influence of HMD motion, thereby improving system stability and control accuracy.

Ultra-Fast Model Predictive Controller (MPC) — 2024.12–2025.12
Advisors: Prof. Gang Chen — *Sun Yat-sen University*

- **Introduction**: Model Predictive Control (MPC) is a robust control method, but its high computational cost limits real-time deployment. We propose a hardware–software co-design approach to accelerate MPC and validate it through UAV experiments.
- **Involvement**:
  - Modeled UAV dynamics and verified state-space equations via MATLAB simulations.
  - Programmed the MPC on an FPGA and controlled UAV flight through ROS + Gazebo simulation.
  - Deployed a hardware-software co-accelerated MPC on a UAV and tested its performance.

Single-cycle CPU and 5-stage pipelined CPU (MIPS) — 2024.12–2025.01
Advisors: Prof. Zhi Zhou — *Sun Yat-sen University*

- **Introduction**: Designed and implemented a single-cycle CPU and a 5-stage pipelined CPU based on the MIPS architecture, covering the complete workflow from hardware design and verification to FPGA deployment.

- **Involvement**:
  - Designed a single-cycle CPU and a 5-stage pipelined CPU, with all hazards (data hazard, structural hazard, and control hazard) resolved in hardware.
  - Implemented both CPUs in Verilog, simulated and debugged them
  - Deployed CPUs on an FPGA development board, respectively.

Operating System Based on the RISC-V Instruction Set 2025.04–2025.08
Advisors: Prof. Pengfei Chen *Sun Yat-sen University*

- **Introduction**: Developed an OS on RISC-V, supporting up to 90 system calls; participated in the Operating System Design Competition.
- **Involvement**:
  - Participated in the overall architecture design of the operating system.
  - Implemented most process management related system calls.
  - Adapted and debugged low-level drivers.

AWARDS & HONORS

Honors

- **2024 National Scholarship**
- **2025 National Scholarship**
- **2024 First Class Academic Scholarship**
- **2025 First Class Academic Scholarship**

Competition Awards

• China Undergraduate Operating System Design Competition	**National Third Prize**	2025.08
• China Undergraduate Mathematics Competition	**Provincial First Prize**	2024.11
• Shenzhen Cup Mathematical Modeling Competition	**Provincial Second Prize**	2024.12
• China Undergraduate Mathematical Modeling Competition	**Provincial Third Prize**	2024.10
• Mathematical Contest in Modeling, USA	**H Award**	2025.05
• MathorCup Mathematical Modeling Competition	**Provincial Second Prize**	2024.04

SKILLS

English:	CET-4 (621), CET-6 (556)
Programming:	Python, C/C++, Matlab, Rust, Verilog
Software:	Linux, Mujoco, ROS, Git
Traits:	Innovative; diligent and detail-oriented; self-learner; persistent and hardworking